

PAPER_201: UQFF Gravitational Waves — Chirp Mass, QNM, BZ, Orbital Decay, and Kilonova

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Session: 50 — grokshare7514fe.txt Full Audit

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Version: 1.0 **Date:** March 13, 2026 **Session:** 50 — grokshare7514fe.txt Full Audit **Author:** Star-Magic UQFF Research Framework **Source:** grokshare7514fe.txt lines 6060–6080 (BBCEquations_04Sept2025.pdf items 1292–1300)

$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot \bigl(1 - U_{bi}/F_U\bigr) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

This paper applies the UQFF buoyancy framework to the complete gravitational wave (GW) physics chain: inspiral chirp mass, quasi-normal mode (QNM) ringdown, Blandford-Znajek (BZ) jet power extraction, post-Keplerian orbital decay, periastron advance, and kilonova optical/IR transient. The F_{UBii}/U_m framework unifies these into a single UQFF operator chain: inspiral ? plunge ? ringdown ? jet ? remnant (kilonova). Numerical coefficients from LIGO GWTC-4.0 and EHT observations calibrate the constants.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. GW inspiral: Chirp Mass

$$F_{UBii, \text{chirp}} = F_{\text{rel}} \times (\kappa / E_{\text{LEP}}) \times Q_{\text{wave}} \times [dE/dt = -(32/5)G^4\mu^2M^3/(c^5r^5)]$$

$$\kappa = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5} \quad (\text{chirp mass, solar masses})$$

$$\kappa = (c^3/G) \cdot (5/96 \cdot p^{-8/3} \cdot f^{-11/3} \cdot \kappa)^{3/5} \quad (\text{from observed GW frequency drift})$$

$$U_m, \text{chirp}(f) = \mu(\kappa_{\text{vac}}) \cdot (1 - e^{-\kappa t}) \cdot (32/5 \cdot G^4 \cdot \mu^2 M^3 / c^5 r^5)$$

Calibration: GW150914 $\kappa \sim 28.3 \text{ M}_\odot$; GW170817 $\kappa \sim 1.188 \text{ M}_\odot$

2. QNM Quasi-Normal Mode Ringdown

$$f_{\text{QNM}} = c^3 / (2pGM_f) \cdot f(a_f)$$

where the Berti et al. fitting function:

$$f_{\text{QNM}} = (c^3 / 2pGM_f) \cdot [0.3737 + 0.088 \cdot a_f + \dots] \quad (l=2, m=2 \text{ dominant mode})$$

Decay time:

$$t_{\text{QNM}} = GM_f / c^3 \cdot g(a_f) \quad (g \sim 2-10 \text{ M for } a_f \sim 0.69)$$

$$F_{UBii, \text{qnm}} = -F_{\text{rel}} \times (f_{\text{QNM}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t_{\text{QNM}}}$$

$$U_m, \text{qnm}(a) = \mu(\kappa_{\text{vac}}) \cdot a \cdot c^3 / (2GM) \times (1 - e^{-\kappa t}) \cdot [l=2 \text{ s}=-2 \text{ perturbation}]$$

$U_{m,damp}(a) = \mu(\tau_{vac}) \cdot Q_{factor} \times (1 - e^{-\tau}) \cdot [Q \sim 10 \text{ for dominant mode}]$
 Calibration: GW150914 final BH $M_f \sim 62 M_\odot$, $a_f \sim 0.67$? $f_{QNM} \sim 251 \text{ Hz}$

3. Blandford-Znajek Jet Power

$P_{BZ} = (1/32) \cdot B^2 \cdot R_{H4} \cdot \Omega_H^2 / c$ (BZ original form)
 Updated EHT form:
 $P_{BZ,EHT} = (\tau/16\pi) \cdot F_{BH}^2 \cdot \Omega_{BH}^2 / c$
 where:
 $\tau \sim 0.044$ (numerical factor from GRMHD)
 $F_{BH} = B \cdot \pi \cdot r_H^2$ (BH magnetic flux, EHT M87* calibrated)
 $\Omega_{BH} = a \cdot c^2 / (2r_H \cdot c)$ (angular velocity)
 For M87*: $B \sim 1\text{--}30 \text{ G}$? $P_{BZ} \sim 10^4 \text{--} 10^5 \text{ erg/s}$
 $F_{UBii,bz} = F_{rel} \times ((1/32) \cdot B^2 \cdot R_{H4} \cdot \Omega_H^2 / c / E_{LEP}) \times Q_{wave}$
 $F_{UBii,bz2} = F_{rel} \times ((\tau/16\pi) \cdot F_{BH}^2 \cdot \Omega_{BH}^2 / c / E_{LEP}) \times Q_{wave}$
 $U_{m,bz}(a) = \mu(\tau_{vac}) \cdot Power \sim B^2 \Omega_H^2 \cdot R_{H4} \times (1 - e^{-\tau})$
 $U_{m,bz2}(a) = \mu(\tau_{vac}) \cdot (\tau/16\pi) F_{BH}^2 \cdot \Omega_{BH}^2 / c \times (1 - e^{-\tau})$

4. Post-Keplerian Orbital Decay

GW orbital decay rate (Peters formula, GR 2.5PN):
 $\tau_b = -(192\pi/5) \cdot (P_b/2\pi)^{-5/3} \cdot (G^3 m_1 m_2 / c^3)^{5/3} / (P_b)^{5/3} \cdot f(e)$
 $f(e) = [1 + (73/24)e^2 + (37/96)e^4] \cdot (1 - e^2)^{-7/2}$
 $F_{UBii,orbdec} = -F_{rel} \times (\tau_b/P_b = dE/dt / E_{LEP}) \times Q_{wave}$
 $U_{m,orbdec}(e) = \mu(\tau_{vac}) \cdot [-dE_{GW}/dt] \times (1 - e^{-\tau})$
 Calibration: Hulse-Taylor PSR B1913+16
 $\tau_{b,obs} \sim -2.422 \times 10^{-12}$ (dimensionless)
 $\tau_{b,GR}$ prediction fits to $<0.1\%$

5. Post-Keplerian Periastron Advance

$\tau_p = 3 \cdot (P_b/2\pi)^{-5/3} \cdot (G(m_1+m_2)/c^3)^{2/3} / (1 - e^2)$
 $F_{UBii,peri} = F_{rel} \times (\tau_p / E_{LEP}) \times Q_{wave} \times (G(m_1+m_2))^{2/3} \cdot (1 - e^2)^{-1}$
 $U_{m,peri}(a) = \mu(\tau_{vac}) \cdot (Kepler: a^3/P^2 = GM/(4\pi^2)) \times (1 - e^{-\tau})$
 Calibration: PSR B1913+16 ? $\tau_p = 4.226^\circ/\text{yr}$ (measured) vs $4.226^\circ/\text{yr}$ (GR)

6. Kilonova Optical/IR Transient

$L_{peak} \sim 10^{41} \cdot (M_{ej}/0.01 M_\odot) \cdot (v_{ej}/0.1c) \cdot (\tau/1 \text{ cm}^2/\text{g})^{-1} \text{ erg/s}$
 Peak timescale: $t_{peak} \sim v(3M_{ej}/(4\pi c v_{ej}))$
 $F_{UBii,kilo} = F_{rel} \times (M_{ej} \cdot v_{ej} \cdot c / (\tau \cdot L_{peak}) / E_{LEP}) \times Q_{wave}$
 $U_{m,kilo}(t) = \mu(\tau_{vac}) \cdot (Diffusion t_d^2 = 3M/(4\pi c v_{ej}^2)) \times (1 - e^{-\tau})$
 Calibration: AT2017gfo (GW170817 neutron star merger):
 $M_{ej} \sim 0.05 M_\odot$, $v_{ej} \sim 0.15c$, $\tau \sim 1\text{--}5 \text{ cm}^2/\text{g}$
 $L_{peak} \sim \text{few} \times 10^{41} \text{ erg/s}$ (r-process nucleosynthesis powered)

7. UQFF GW Chain Unification

The entire GW compact binary lifecycle maps onto UQFF operators:

1. Inspiral ? F_UBii,chirp + Um,chirp [orbital energy loss, GW frequency sweep]
? plunge/merger
2. Ringdown ? F_UBii,qnm + Um,qnm [ringing BH, quenches exponentially]
? spin-down
3. Jet launch ? F_UBii,bz + Um,bz [magnetically extracted power]
? r-process
4. Remnant ? F_UBii,kilo + Um,kilo [neutron star ejecta optical transient]
Long-term:
5. Orbital decay ? F_UBii,orbdec + Um,orbdec [binary evolving toward merger]
6. Periastron ? F_UBii,peri + Um,peri [GR precession measured]

8. Numerical Summary

Process	UQFF Parameter	Calibration System
Chirp mass ??	28.3 M_?	GW150914
QNM freq	251 Hz	GW150914
BZ power	104 ² ?4 ³ erg/s	M87* EHT
?b/?b,GR	<0.1% deviation	PSR B1913+16
?? match	4.226°/yr	PSR B1913+16
Kilonova L	few×104 ¹ erg/s	AT2017gfo

9. References

grok_share_7514fe.txt lines 6060–6080 (BBCEquations items 1292–1300, 1556–1560)

PAPER198: FUBii Taxonomy Part 1 (QNM/BZ/Sedov)

PAPER_200: Um Universal Magnetism Catalogue

LIGO GWTC-4.0 (2025 catalog)

EHT M87* 2019 polarization papers